Matthew Tan W2

Q1. (a) i. $\vdash_K \Box(\phi \to \psi) \to (\Box\phi \to \Box\psi)$

Assume for reductio, that there is some K-model, M s.t.

$$V_M(\Box(\phi \to \psi) \to (\Box\phi \to \Box\psi), w) = 0$$

Then,

$$V_M(\Box(\phi \to \psi), w) = 1 \quad ①$$

and

$$V_M(\Box\phi \to \Box\psi, w) = 0 \quad ②$$

From ①: $V_M(\phi \to \psi, v) = 1$, for all $v \in W$ s.t. Rwv

$\therefore V_M(\phi, v) = 0$ or $V_M(\psi, v) = 1$ for all $v \in W$ s.t. Rwv
(a) (b)

From ②: $V_M(\Box\phi, w) = 1$ and $V_M(\Box\psi, w) = 0$

$\therefore V_M(\phi, v) = 1$ for all $v \in W$ s.t. Rwv

and $V_M(\psi, v) = 0$ for some $v \in W$ s.t. Rwv

✓

But then if (a), there is some v s.t.

$$V_M(\psi, v) = 0 \text{ and } V_M(\psi, v) = 1 \quad \times$$

But if (b), there is some v s.t.

$$V_M(\phi, v) = 0 \text{ and } V_M(\phi, v) = 1 \quad \times$$

contradiction by dichotomy

ii. $\vDash_D \Box\phi \to \Diamond\phi$

Assume for reductio that there is some D-model, M, s.t.

$V_M(\Box\phi \to \Diamond\phi, w) = 0$, for some w (i.e. not valid in the M-model)

$\therefore V_M(\Box\phi, w) = 1$ and $V_M(\Diamond\phi, w) = 0$

which implies

$V_M(\phi, v) = 1$ for all v s.t. Rwv ①

and $V_M(\phi, v) = 0$ for all v s.t. Rwv ②

Since R is serial on W, we know that for any w, there is at least one world, call this u, accessible from w.

$\therefore$ From ①, ②, there is a u s.t. Rwu and $V_M(\phi, u) = 1$ and $V_M(\phi, u) = 0$ X

✓

iii. $\vDash_T \Box\phi \to \phi$

Assume for reductio, that there is a T-model s.t.

$V_T(\Box\phi \to \phi, w) = 0$ for some $w \in W$

Then,

$V_T(\Box\phi, w) = 1$ for that world w ①

and

$V_T(\phi, w) = 0$ for that world w ②

This means that. $V_T(\phi, v) = 1$ for all Rwv

Due to reflexivity, we know that in any T model, Rww. So from ①; then $V_T(\phi, w) = 0$ for w s.t. Rww

$\therefore$ So, there is some w s.t. $V_T(\phi, w) = 1$ and $V_T(\phi, w) = 0$

✓

iv. $\vDash_B \phi \to \Box\Diamond\phi$

AFC that there is a B-model, $\mathcal{M}$,

s.t. $V_M(\phi \to \Box\Diamond\phi, w) = 0$ for some $w \in W$

Then, $V_M(\phi, w) = 1$ ①

and $V_M(\Box\Diamond\phi, w) = 0$ ②

From ②, $V_M(\Diamond\phi, v) = 0$ for some v s.t. Rwv

Since R is symmetric, if $V_M(\Diamond\phi, v) = 0$ for some v s.t. Rwv

then $V_M(\Diamond\phi, w) = 0$ for some v s.t. Rvw

$\therefore V_M(\phi, u) = 0$ for all u s.t. Rwu

no, this move requires transitivity

Since R is reflexive, then we can say $u = w$,

s.t. $V_M(\phi, w) = 0$ which contradicts ① $\times$

v. $\vDash_{S4} \Box\phi \to \Box\Box\phi$

let $\mathcal{M}$ be some S-model s.t.

$V_M(\Box\phi \to \Box\Box\phi, w) = 0$

Then, $\underset{①}{V_M(\Box\phi, w) = 1}$ and $\underset{②}{V_M(\Box\Box\phi, w) = 0}$

$\therefore$ From ①, $V_M(\phi, v) = 1$ for all v s.t. Rwv ③

From ②, $V_M(\Box\phi, u) = 0$ for some u s.t. Rwu

$\therefore V_M(\phi, u') = 0$ for some u' s.t. Ruu'

Due to transitivity, Rwu and Ruu' implies Rwu'

Then, from ③, we can pick a $v = u'$ s.t.

$V_M(\phi, u') = 1$ for that u' s.t. Rwu' ✗

vi. $\models_{S5} \Diamond\phi \to \Box\Diamond\phi$

AFC that there is an S5-model, M, s.t.

$V_M(\Diamond\phi \to \Box\Diamond\phi, w) = 0$ for some w

Then,

$V_M(\Diamond\phi, w) = 1$ ①

and $V_M(\Box\Diamond\phi, w) = 0$ ②

From ①, there is some v s.t. Rwv and

$V_M(\phi, v) = 1$

From ②, $V_M(\Diamond\phi, u) = 0$ for some u s.t. Rwu

$\therefore V_M(\phi, u') = 0$ for all u' s.t. Ruu'

Due to symmetry, Rwu implies Ruw

$\therefore V_M(\phi, w) = 0$

Due to reflexivity, Rww, so
$V_M(\phi, w) = 1$ $\times$

This does not follow

$\times$

(b) i. $\nvDash_K \Box\phi \to \Diamond\phi$

So, there is some model M for which

$$V_M(\Box\phi \to \Diamond\phi, w) = 0 \text{ for some } w$$

So $V_M(\Box\phi, w) = 1$

$\therefore V_M(\phi, v) = 1$ for all v s.t. Rwv

and $V_M(\Diamond\phi, w) = 0$

$\therefore V_M(\phi, u) = 0$ for all u s.t. Rwu

$$\boxed{\begin{array}{ll} r & \underset{1}{\Box\phi} \underset{0}{\to} \underset{0}{\Diamond\phi} \end{array}}$$

Counter model is

$W = \{r\}$

$R = \emptyset$ (since no world is accessible from r)

$V_M(\phi, w) = 1$

✓

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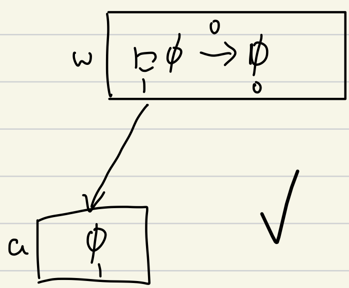
ii. $\nvDash_D \Box\phi \to \phi$

there is some M s.t.

$V_M(\Box\phi \to \phi, w) = 0$

$V_M(\Box\phi, w) = 1$ and $V(\phi, w) = 0$

$V_M(\phi, v) = 1$ for all v s.t. Rwv



So, the countermodel is

$W = \{w, a\}$

$R = \{\langle w, a\rangle\}$

$V_M(\phi, r) = 0$

$V_M(\phi, a) = 1$

iii. $\nvDash_{S4} \phi \to \Box\Diamond\phi$

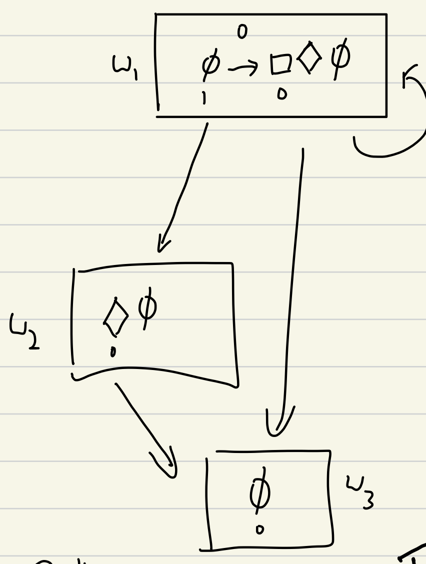
$\exists M$ s.t. $V_M(\phi \to \Box\Diamond\phi, w) = 0$

$\therefore V_M(\phi, w) = 1$

$V_M(\Box\Diamond\phi, w) = 0$

$\therefore V_M(\Diamond\phi, v) = 0$ for some v s.t. Rwv

$\therefore V_M(\phi, u) = 0$ for all u s.t. Rvu



So, the countr model is

$W = \{w_1, w_2, w_3\}$

$R = \{\langle w_1, w_1\rangle, \langle w_1, w_2\rangle, \langle w_2, w_3\rangle\}$

$V_M(\phi, w_1) = 1$, $V_M(\phi, w_2) = 0$, $V_M(\phi, w_3) = 0$

wb is not reflexive

This is not S4, in particular

iv. $\nvDash_B \Box\phi \to \Box\Box\phi$

$\exists M$ s.t. $V_M(\Box\phi \to \Box\Box\phi, w) = 0$

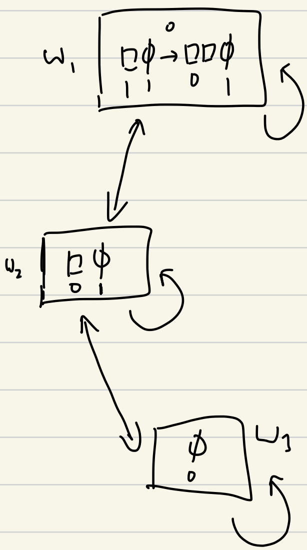
$V_M(\Box\phi, w) = 1$

$\therefore V_M(\phi, v) = 1$ for all v s.t. Rwv

$V_M(\Box\Box\phi, w) = 0$

$\therefore V_M(\Box\phi, u) = 0$ for some u s.t. Rwu

$\therefore V_M(\phi, u') = 0$ for some u' s.t. Ruu'



so, $W = \{w_1, w_2, w_3\}$

$R = \{\langle w_1, w_1\rangle, \langle w_2, w_2\rangle, \langle w_3, w_3\rangle, \langle w_1, w_2\rangle, \langle w_2, w_1\rangle, \langle w_2, w_3\rangle, \langle w_3, w_2\rangle\}$

$V_M(\phi, w_1) = 1$, $V_M(\phi, w_2) = 1$, $V_M(\phi, w_3) = 0$ ✓

v. $\nvDash_{S4} \Diamond\phi \to \Box\Diamond\phi$ and $\nvDash_{B} \Diamond\phi \to \Box\Diamond\phi$

let M be a model s.t.

$V_M(\Diamond\phi \to \Box\Diamond\phi, w) = 0$

so, $V_M(\Diamond\phi, w) = 1$

$\therefore V_M(\phi, v) = 1$ for some v s.t. Rwv ①

and $V_M(\Box\Diamond\phi, w) = 0$

$\therefore V_M(\Diamond\phi, u) = 0$ for some u s.t. Rwu ②

$\therefore V_M(\phi, u') = 0$ for all u' s.t. Ruu' ③

A countermodel for S5 is not possible → That was 1a) vi

we want to show that all worlds accessible from w

is also accessible from u

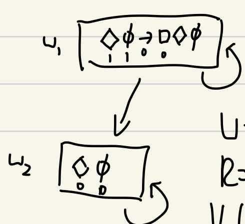
Assume for reductio that $\exists v$ s.t. Rwv but not Ruv

But by symmetry, Rwu implies Ruw. And by transitivity,

Ruw and Rwv implies Ruv ✗

So, must look for indiv S4 and B models

S4 counter model:

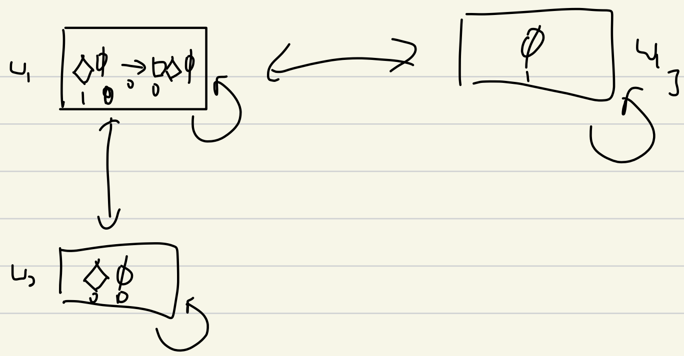


$W = \{w_1, w_2\}$

$R = \{\langle w_1, w_1\rangle, \langle w_2, w_2\rangle, \langle w_1, w_2\rangle\}$

$V_M(\phi, w_1) = 1, V_M(\phi, w_2) = 0$

B - countermodel:



$\therefore$ $W = \{w_1, w_2, w_3\}$

$R = \{\langle w_1, w_1\rangle, \langle w_2, w_2\rangle, \langle w_3, w_3\rangle, \langle w_1, w_2\rangle, \langle w_2, w_1\rangle,$
$\langle w_1, w_3\rangle, \langle w_3, w_1\rangle\}$

$V_M(\phi, w_1) = 0, \; V_M(\phi, w_2) = 0$ ✓

$V_M(\phi, w_3) = 1$

Q2. (a) i. $\models_{PTL} \phi \to HF\phi$

Suppose that $\exists M$ s.t.

$V_M(\phi \to HF\phi, t^*) = 0$

Then $V_M(\phi, t) = 1$ (✱)

and $V_M(HF\phi, t) = 0 \Rightarrow V_M(F\phi, t') = 0$ for some $t' \leq t$

$\therefore V_M(\phi, t'') = 0$ for all $t'' \geq t'$

Notice that no matter what t' is (either $t' = t$ or $t' < t$),

since $V_M(\phi, t'') = 0$ for all $t'' \geq t'$,

this means that $V_M(\phi, t) = 0$, contradicting (*)

ii. $PP\phi \to P\phi$ is not valid

$\nvDash_{PTL} PP\phi \to P\phi$

Let $V_M(PP\phi \to P\phi, t) = 0$

then $V_M(PP\phi, t) = 1$ ①

and $V_M(P\phi, t) = 0$ ②

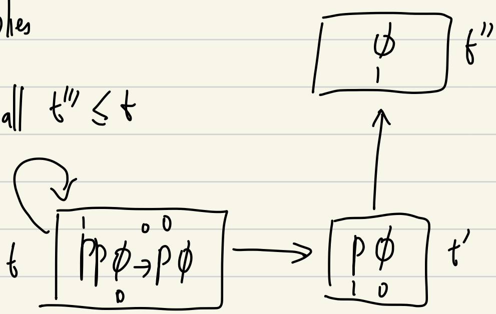
From ①, $V_M(P\phi, t') = 1$ for some $t' \leq t$

$\therefore V_M(\phi, t'') = 1$ for some $t'' \leq t'$

From ②, $V_M(P\phi, t) = 0$ implies

$V_M(\phi, t''') = 0$ for all $t''' \leq t$

Then construct a counter model



so, $W = \{t, t', t''\}$, $R = \{\langle t, t' \rangle, \langle t', t'' \rangle\}$, $V_M(\phi, t) = 0$, $V_M(\phi, t') = 0$

$V_M(\phi, t'') = 1$

iii. $(F\phi \wedge F\psi) \rightarrow (F(\phi \wedge F\psi) \vee F(\psi \wedge F\phi) \vee F(\phi \wedge \psi))$

let $V_M((F\phi \wedge F\psi) \rightarrow (F(\phi \wedge F\psi) \vee F(\psi \wedge F\phi) \vee F(\phi \wedge \psi), t_1) = 0$

Then $V_M(F\phi \wedge F\psi, t_1) = 1$

$\therefore V_M(F\phi, t_1) = 1$ and $V_M(F\psi, t_1) = 1$

$\therefore V_M(\phi, t_2) = 1$ for some $t_2 \geq t_1$ ①

$V_M(\psi, t_3) = 1$ for some $t_3 \geq t_1$ ②

Then, $V_M(F(\phi \wedge F\psi) \vee F(\psi \wedge F\phi) \vee F(\phi \wedge \psi), t_1) = 0$

implies $\underset{(a)}{V_M(F(\phi \wedge F\psi), t_1)} = \underset{(b)}{V_M(F(\psi \wedge F\phi), t_1)} = \underset{(c)}{V_M(F(\phi \wedge \psi), t_1)} = 0$

From (a), $V_M(\phi \wedge F\psi, t_4) = 0$ for all $t_4 \geq t_1$

for all $t_4 \geq t_1$, (either $V(\phi, t_4) = 0$

or $V(F\psi, t_4) = 0$

$\therefore V(\psi, t_5) = 0$ for all $t_5 \geq t_4$)

From (b), $V_M(\psi \wedge F\phi, t_6) = 0$ for all $t_6 \geq t_1$

So, for all $t_6 \geq t_1$ ($V_M(\psi, t_6) = 0$ or

$V_M(F\phi, t_6) = 0 \Rightarrow V(\phi, t_7) = 0$ for all $t_7 \geq t_6$)

From (c), $V_M(\phi \wedge \psi, t_8) = 0$ for all $t_{10} \geq t_1$,

then for all $t_8 \geq t_1$, $V_M(\phi, t_8) = 0$ or $V_M(\psi, t_8) = 0$

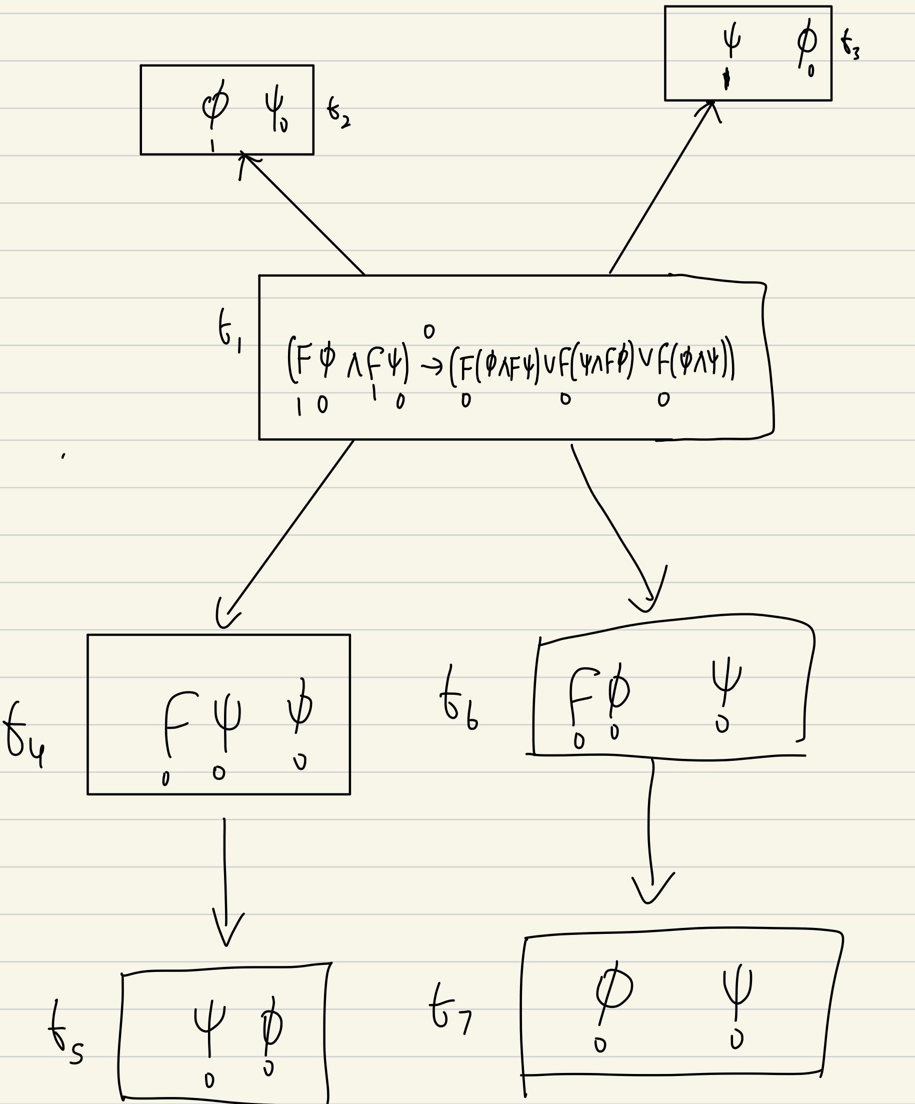
Given these constraints, consider a countermodel

In summary: (1) ϕ, ψ must be true in one world accessible from t_1 (but not the same world)

(2) $F\phi$, $F\psi$ must be true in t_1 but false in all worlds accessible from it

(3) ϕ, ψ can never be true in worlds accessible from worlds where $F\phi$, $F\psi$ are false

(4) ϕ, ψ can never be both true in a world accessible from t_1



$W = \{t_1, t_2, t_3, t_4, t_5, t_6, t_7\}$, $R = \{\langle t_1, t_2\rangle, \langle t_1, t_3\rangle, \langle t_1, t_4\rangle, \langle t_1, t_5\rangle, \langle t_5, t_6\rangle, \langle t_5, t_7\rangle\}$

Important valuations:

$$V_M(\phi, t_2) = V_M(\psi, t_3) = 1$$

$$V_M(\phi, t_3) = V_M(\psi, t_2) = 0$$

$$V_M(\phi, t_4) = V_M(\phi, t_5) = V_M(\phi, t_6) = V_M(\phi, t_7)$$

$$= V_M(\psi, t_4) = V_M(\psi, t_5) = V_M(\psi, t_6) = V_M(\psi, t_7) = 0$$

(b) We can impose a constraint of transitivity

so that $\models_{PTL} PP\phi \to P\phi$

AFC $V_M(PP\phi \to P\phi, t_1) = 0$

$V_M(PP\phi, t_1) = 1$ ①

and $V_M(P\phi, t_1) = 0$ ②

From ①, $V_M(P\phi, t_2) = 1$ for some $t_2 \leq t_1$

$V_M(\phi, t_3) = 1$ for some $t_3 \leq t_2$

From ②, $V_M(\phi, t_4) = 0$ for all $t_4 \leq t_1$

since $t_2 \leq t_1$ and $t_3 \leq t_2$ imply $t_3 \leq t_1$

by transitivity, there is some time t'

s.t. $V_M(\phi, t') = 1$ and $V_M(\phi, t') = 0$ ※

We can also impose a constraint of '$\leq$' being a linear order to make

$$\vDash_{PTL} (F\phi \wedge F\psi) \rightarrow (F(\phi \wedge F\psi) \vee F(\psi \wedge F\phi) \vee F(\phi \wedge \psi))$$

let $V_M((F\phi \wedge F\psi) \rightarrow (F(\phi \wedge F\psi) \vee F(\psi \wedge F\phi) \vee F(\phi \wedge \psi), t_1) = 0$

Then $V_M(F\phi \wedge F\psi, t_1) = 1$

$\therefore V_M(F\phi, t_1) = 1$ and $V_M(F\psi, t_1) = 1$ ✓

$\therefore V_M(\phi, t_2) = 1$ for some $t_2 \geq t_1$ ①

$V_M(\psi, t_3) = 1$ for some $t_3 \geq t_1$ ②

Then, $V_M(F(\phi \wedge F\psi) \vee F(\psi \wedge F\phi) \vee F(\phi \wedge \psi), t_1) = 0$

implies $\underset{(a)}{V_M(F(\phi \wedge F\psi), t_1)} = \underset{(b)}{V_M(F(\psi \wedge F\phi), t_1)} = \underset{(c)}{V_M(F(\phi \wedge \psi), t_1)} = 0$

From (a), $V_M(\phi \wedge F\psi, t_4) = 0$ for all $t_4 \geq t_1$

for all $t_4 \geq t_1$, (either $V(\phi, t_4) = 0$

or $V(F\psi, t_4) = 0$

$\therefore V(\psi, t_5) = 0$ for all $t_5 \geq t_4$)

From (b), $V_M(\psi \wedge F\phi, t_6) = 0$ for all $t_6 \geq t_1$

So, for all $t_6 \geq t_1$ ($V_M(\psi, t_6) = 0$ or

$V_M(F\phi, t_6) = 0 \Rightarrow V(\phi, t_7) = 0$ for all $t_7 \geq t_6$)

From (c), $V_M(\phi \wedge \psi, t_8) = 0$ for all $t_{10} \geq t_1$,

then for all $t_8 \geq t_1$, $V_M(\phi, t_8) = 0$ or $V_M(\psi, t_8) = 0$

Thus, to show a contradiction, we have to prove by trichotomy.

Case 1: $t_2 = t_3$

Then, it clearly contradicts (c) since

there is some t' where $t' \geq t_1$, $V_M(\phi, t') = V_M(\psi, t') = 1$

Case 2: $t_2 < t_3$, which contradicts (a)

Suppose at t_2, $V_M(\phi, t_2) = 1$

Then $V_M(\psi, t'') = 0$ for all $t'' \geq t_2$

but by assumption, $V_M(\psi, t_3) = 1$ for some $t_3 \geq t_2$

By strong connectedness, for any t'', either $t_3 \geq t''$ or $t'' \geq t_3$.

Notice that we can choose a $t'' = t_3$ which satisfies strong connectedness due to reflexivity & antisymmetry of $\geq$

$\therefore V_M(\psi, t'') = 1$ and $V(\psi, t'') = 0$

Case 3: $t_3 < t_2$, which contradicts (b)

Suppose at t_3, $V_M(\psi, t_3) = 1$

Then $V_M(\phi, t''') = 0$ for all $t''' \geq t_3$

but by assumption, $V_M(\phi, t_2) = 1$ for some $t_2 \geq t_3$

By strong connectedness & anti-symmetry,
we can choose a $t'' = t_2$ that satisfies
either $t'' \geq t_2$ or $t_2 \geq t''$

$\therefore V_M(\phi, t'') = 1$ and $V_M(\phi, t'') = 0$ ✗

Q3 (a) It is either possible that ϕ and possible that not ϕ can be said shorter

(b) $V_M(C\phi, w) = 1$ iff $V_M(\Diamond\phi \wedge \Diamond\sim\phi, w) = 1$

iff $V_M(\Diamond\phi, w) = 1$ and $V(\Diamond\sim\phi, w) = 1$

iff $V_M(\phi, u) = 1$ for some u s.t. Rwu

and $V_M(\sim\phi, u') = 1$ for some u' s.t. Rwu'

iff $V_M(\phi, u) = 1$ for some u s.t. Rwu
and $V_M(\phi, u') = 0$ for some u' s.t. Rwu'

(c) i. $\vDash_{S5} C\phi \to \sim CC\phi$

AFC that $V_M(C\phi \to \sim CC\phi, w) = 0$

$\therefore V_M(C\phi, w) = 1$ ①

and $V_M(\sim CC\phi, w) = 0 \Rightarrow V_M(CC\phi, w) = 1$ ②

from ②

$V_M(C\phi, v) = 1$ for some v s.t. Rwv

and $V_M(C\phi, v') = 0$ for some v' s.t. Rwv'

$\therefore V_M(\phi, u) = 1$ for some u s.t. Rvu

and $V_M(\phi, u') = 0$ for some u' s.t. Rvu'

$\therefore V_M(\phi, a) = 0$ for all a s.t. $Rv'a$ (a)

or $V_M(\phi, a') = 1$ for all a' s.t. $Rv'a'$ (b)

from ①, $V_M(\phi, b) = 1$ for some b s.t. Rwb

$V_M(\phi, b') = 0$ for some b' s.t. Rwb'

By dichotomy, assume (b)

since $Rwv' \Rightarrow Rv'w$.

$Rv'w$ and Rwb' imply $Rv'b'$

so, there is some world where

$V_M(\phi, b') = 1$ and $V_M(\phi, b') = 0$ $\times$

assume (a),

since $Rwv' \Rightarrow Rv'w$ ✓

$Rv'w$ and $Rwb \Rightarrow Rv'b$

so there is some world accessible to v'

$V_M(\phi, b) = 1$ and $V_M(\phi, b) = 0$ $\times$

ii. $\vDash_{S5} {\sim}C\phi \rightarrow {\sim}C{\sim}C\phi$

AFC that $V_M({\sim}C\phi \rightarrow {\sim}C{\sim}C\phi, w_1) = 0$

Then, $V_M({\sim}C\phi, w_1) = 1$ and $V_M({\sim}C{\sim}C\phi, w_1) = 0$

$\therefore V_M(C\phi, w_1) = 0$ ① $\therefore V_M(C{\sim}C\phi, w_1) = 1$ ②

From ①, $V_M(\phi, w_2) = 0$ for all w_2 s.t. Rw_1w_2 ⓐ

or $V_M(\phi, w_3) = 1$ for all w_3 s.t. Rw_1w_3 ⓑ

From ②, $V_M(\sim C\phi, w_4) = 1$ for some w_4 s.t. Rw_1w_4

and $V_M(\sim C\phi, w_5) = 0$ for some w_5 s.t. Rw_1w_5

$\therefore V_M(C\phi, w_4) = 0$ for some w_4 s.t. Rw_1w_4

and $V_M(C\phi, w_5) = 1$ for some w_5 s.t. Rw_1w_5

$\therefore (V_M(\phi, w_6) = 0$ for all w_6 s.t. Rw_4w_6

or $V_M(\phi, w_7) = 1$ for all w_7 s.t. $Rw_4w_7)$

and $(V_M(\phi, w_8) = 1$ for some w_8 s.t. Rw_5w_8 and

$V_M(\phi, w_9) = 0$ for some w_9 s.t. $Rw_5w_9)$

By dichotomy, say (a) holds.

$Rw_1w_5 \;\&\; Rw_5w_8 \Rightarrow Rw_1w_8$

from (a), this means that at some w_8,

$$V_M(\phi, w_8)=1 \text{ and } V_M(\phi, w_8)=0$$

Say (b) holds,

$$Rw_1w_5 \; \& \; Rw_5w_9 \Rightarrow Rw_1w_9$$

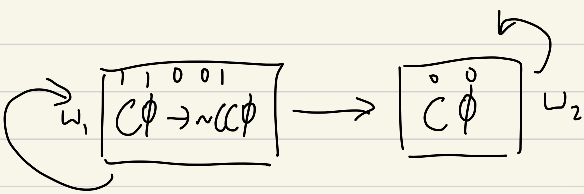
Since (b) says $V_M(\phi, w_3)=1$ for all w_3 s.t. Rw_1w_3

Then, at some w_9,

$$V_M(\phi, w_9)=1 \text{ and } V_M(\phi, w_9)=0$$

$\Rightarrow$ contradiction established by dichotomy ✓

(d) if we replace S5 with S4, no longer symmetric



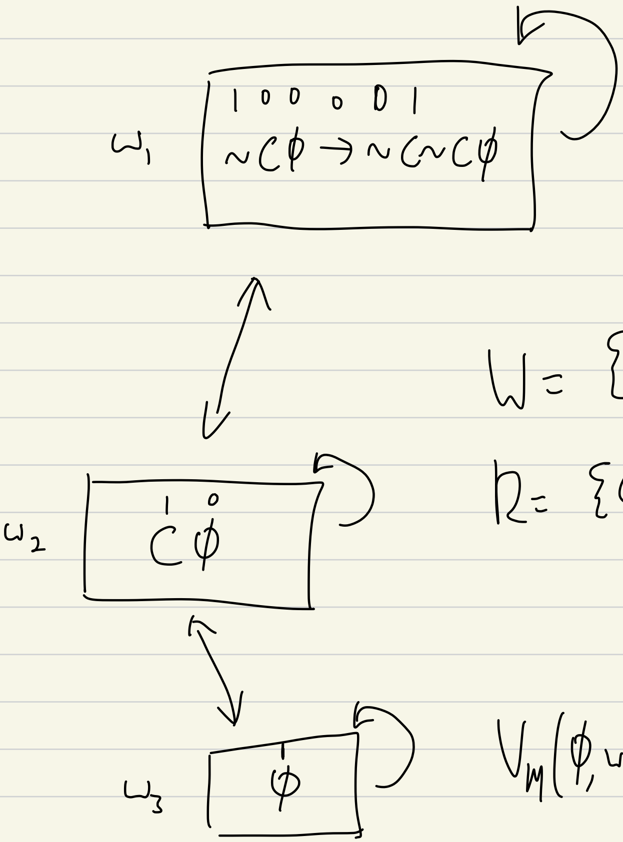
✓

$W=\{w_1, w_2\}$ $\quad R=\{\langle w_1,w_1\rangle, \langle w_2,w_2\rangle, \langle w_1,w_2\rangle\}$

$$V_M(\phi, w_1)=1, \; V_M(\phi, w_2)=0$$

$\therefore \nvDash_{S4} C\phi \to \sim CC\phi$

(e) if we replace S5 with B, no longer transitive



$W = \{w_1, w_2, w_3\}$

$R = \{\langle w_1, w_1\rangle, \langle w_2, w_2\rangle, \langle w_3, w_3\rangle, \langle w_1, w_2\rangle, \langle w_2, w_1\rangle, \langle w_2, w_3\rangle, \langle w_3, w_2\rangle\}$

$V_M(\phi, w_1) = V_M(\phi, w_2) = 0$

$V_M(\phi, w_3) = 1$

$V_M(\sim C\phi, w_1) = 0$ as

$V_M(C\phi, w_1) = 1$ as

$V_M(\phi, w_2) = 0$ and $V_M(\phi, w_3) = 1$

(f) (i) says if it is possible that ϕ and possible that it is not ϕ

then it is not the case that (it is possible that (it is possible that ϕ and possible that not ϕ)

and it is possible that not (it is possible that ϕ and possible that not ϕ))

This does not seem to be obviously valid since the antecedent need only be true relative to some world w_1

and the consequent, namely the second conjunct can be true relative to some other world w_2 accessible from w_1 and hence does not obviously contradict the antecedent

(ii) says that if either ϕ is necessary or ϕ is impossible, then

it is either necessary that (either ϕ is necessary or ϕ is impossible)

or impossible that (either ϕ is necessary or ϕ is impossible)

Again, this is not obviously valid since just because the antecedent is true in w_1 doesn't mean it is true in all worlds accessible from it, and it clearly does not imply that its content is impossible

So, just because (i) & (ii) are valid in S5 but not in S4 & B respectively, does not imply S5 is correct since there is little intuitively valid about (i) & (ii)

Q4. (a) i. Claim: if $O_1 \equiv_s O_2$, then $OO_1 \equiv_s OO_2$

Proof. Assume $O_1 \equiv_s O_2$,

then $\models_s O_1\phi \leftrightarrow O_2\phi$ by def of $\equiv_s$

Since if $\models \psi_1 \leftrightarrow \psi_2$, then $\models \chi(\psi_1) \leftrightarrow \chi(\psi_2)$

we can define $\psi_1 := O_1\phi_1$

$\psi_2 := O_2\phi_2$

and $\chi(\psi) := O\psi$

$\therefore \models_s \chi(O_1\phi_1) \leftrightarrow \chi(O_2\phi_2)$ ✓

$\therefore \models_s OO_1\phi_1 \leftrightarrow OO_2\phi_2$

$\therefore \models_s OO_1 \equiv OO_2$

ii. Claim: if $O_1 \equiv_s O_2$, then $O_1O \equiv_s O_2O$

By assumption, $O_1 \equiv_s O_2$

$\therefore \models_s O_1\phi \leftrightarrow O_2\phi$

then, let $\chi(\cdot)$ be some transformation such that

$\chi(O_i\phi) := O_iO\phi$

This move is not allowed

$\therefore$ Since if $\models_S \psi_1 \leftrightarrow \psi_2$, then $\models_S \mathcal{X}(\psi_1) \leftrightarrow \mathcal{X}(\psi_2)$

then, $\models_S \bigcirc_1 \bigcirc \phi \leftrightarrow \bigcirc_2 \bigcirc \phi$

$\therefore \bigcirc_1 \bigcirc \equiv_S \bigcirc_2 \bigcirc$

(b) i. Claim: There are infinitely many modalities expressed by strings in B

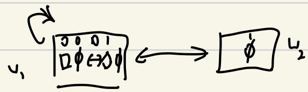
let $C(\phi)$ denote the length of the string i.e $\Box\phi$ has length 1

we make a different claim that for any $C(\phi)=n$, the number of modalities expressed is 2^n for some wff ϕ

BC: $C(\phi)=1$. (since $\bigcirc$ is non-empty)

Then let $\bigcirc_1\phi$ be $\Box\phi$ & $\bigcirc_2\phi$ be $\Diamond\phi$

Trivially, $\not\models_B \Box\phi \leftrightarrow \Diamond\phi$



So, there are 2^1 modalities.
BC verified

$W = \{w_1, w_2\}$ $\qquad V_M(\phi, w_1)=0,\ V_M(\phi, w_2)=1$

$R = \{\langle w_1,w_1\rangle, \langle w_1,w_2\rangle, \langle w_2,w_1\rangle, \langle w_2,w_2\rangle\}$

IH: if $C(\phi)=n$ and there are 2^n modalities for some wff ϕ,

Then there are 2^{n+1} modalities for ψ when $C(\psi)=n+1$

Then, consider either $\Box\bigcirc\phi$ and $\Diamond\bigcirc\phi$

we want to show that they express different modalities

$\not\models_B \Box\bigcirc\phi \leftrightarrow \Diamond\bigcirc\phi$

There is an M s.t. $V_M(\Box\bigcirc\phi \leftrightarrow \Diamond\bigcirc\phi, w)=0$

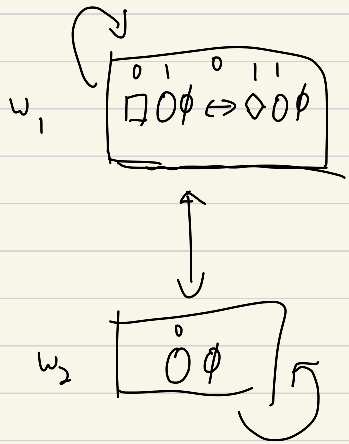
$\therefore V_M(\Box\bigcirc\phi, w) \neq V_M(\Diamond\bigcirc\phi, w)$

if $V_M(\Box O\phi, w) = 0$, $V_M(\Diamond O\phi, w) = 1$

then there is some v s.t. Rwv

where $V_M(O\phi, v) = 0$

and $V_M(O\phi, u) = 1$ for some u s.t. Rwu



so $W = \{w_1, w_2\}$, $R = \{\langle w_1, w_1\rangle, \langle w_2, w_2\rangle, \langle w_1, w_2\rangle, \langle w_2, w_1\rangle\}$

You should be a bit more explicit about the use of your induction hypothesis

$V_M(O\phi, w_1) = 1$, $V_M(O\phi, w_2) = 0$

$\therefore \nvDash_B \Box O\phi \leftrightarrow \Diamond O\phi$

$\therefore \nvDash_B O_1 O\phi \leftrightarrow O_2 O\phi$, $O_1 O \not\equiv_B O_2 O$

by contraposition

$\therefore O_1 \not\equiv_B O_2$

Since O was an arbitrary modality & we know by IH that there are 2^n such Os, O_1 & O_2 each express a unique modality for each 2^n modality

$\therefore$ There are 2^{n+1} modalities for ψ

Claim established

Let $C(\phi) = n$, then there are 2^n modalities for ϕ.

$\lim_{n\to\infty} C(\phi) \to \infty$, $\lim_{n\to\infty} 2^n \to \infty$

$\therefore$ there are infinitely many modalities for strings expressed in B for any ϕ

ii. Claim: there are exactly two modalities expressed by strings in S5

Again we prove this by induction

BC: $C(\phi) = 1$. $\nvDash_{S5} \Box\phi \leftrightarrow \Diamond\phi$

from i. since the B-countermodel respects S5's accessibility relations

so there are 2 modalities

IH: if $C(\phi) = n$ & there are 2 modalities, then there are 2 modalities for ψ where $C(\psi) = n+1$

By IH, this means that for ϕ where $C(\phi) = n$,

for any O, $\vDash_{S5} O\phi \leftrightarrow \Diamond\phi$ or $\vDash_{S5} O\phi \leftrightarrow \Box\phi$

$\therefore$ it suffices to show that for any concatenation $\Box O\phi$ or $\Diamond O\phi$, they are equivalent to either $\Diamond\phi$ or $\Box\phi$

Case 1: let $O\phi \equiv_{S5} \Box\phi$, and we append $\Box$ to $O\phi$

then, we claim that $\models_{S5} \Box\Box\phi \leftrightarrow \Box\phi$

AFC that $V_M(\Box\Box\phi \leftrightarrow \Box\phi, w) = 0$

Case A: $V_M(\Box\Box\phi, w) = 1$, $V_M(\Box\phi, w) = 0$

$\therefore V_M(\phi, v) = 0$ for some v s.t. Rwv

$\therefore V_M(\Box\phi, u) = 1$ for all u s.t. Rwu

$\therefore V_M(\phi, u') = 1$ for all u' s.t. Ruu'

By symmetry, $Rwu \Rightarrow Ruw$. By transitivity,

Ruw & $Rwv \Rightarrow Ruv$

so v is accessible from u

$\therefore V_M(\phi, v) = 1$ & $V_M(\phi, v) = 0$ $\times$

Case B: $V_M(\Box\Box\phi, w) = 0$, $V_M(\Box\phi, w) = 1$

$\therefore V_M(\Box\phi, v) = 0$ for some v s.t. Rwv

$\therefore V_M(\phi, u) = 0$ for some u s.t. Rvu

$\therefore V_M(\phi, u') = 1$ for all u' s.t. Rwu'

By transitivity Rwv & $Rvu \Rightarrow Rwu$

$\therefore V_M(\phi, u) = 0$ & $V_M(\phi, u') = 1$ $\times$

$\therefore \models_{S5} \Box\Box\phi \leftrightarrow \Box\phi$

Case 2: let $O\phi \equiv_{S5} \Box\phi$, and we append $\Diamond$ to $O\phi$

then, claim that $\models_{S5} \Diamond\Box\phi \leftrightarrow \Box\phi$

AFC that $V_M(\Diamond\Box\phi \leftrightarrow \Box\phi, w) = 0$

if $V_M(\Diamond\Box\phi, w) = 1$, $V_M(\Box\phi, w) = 0$

then $V_M(\phi, v) = 0$ for some v s.t. Rwv

$V_M(\Box\phi, u) = 1$ for some u s.t. Rwu

$V_M(\phi, u') = 1$ for all u' s.t. Ruu'

$Rwu \Rightarrow Ruw$

$Ruw \,\&\, Rwv \Rightarrow Ruv$

$\therefore V_M(\phi, v) = 1 \,\&\, V_M(\phi, v) = 0$ ✗

if $V_M(\Diamond\Box\phi, w) = 0$, $V_M(\Box\phi, w) = 1$

$\therefore V_M(\phi, v) = 1$ for all v s.t. Rwv

$\therefore V_M(\Box\phi, u) = 0$ for all u s.t. Rwu

$\therefore V_M(\phi, u') = 0$ for some u' s.t. Ruu'

$Rwu \,\&\, Ruu' \Rightarrow Rwu'$

$\therefore V_M(\phi, u') = 0 \,\&\, V_M(\phi, u') = 1$ ✗

Case 3: let $\bigcirc\phi \equiv_{S5} \Diamond\phi$ & append $\Box$ to $\bigcirc\phi$

claim: $\vDash_{S5} \Box\Diamond\phi \leftrightarrow \Diamond\phi$

AFC that $V_M(\Box\Diamond\phi \leftrightarrow \Diamond\phi, w) = 0$

if $V_M(\Box\Diamond\phi, w) = 1$, $V_M(\Diamond\phi, w) = 0$

$\therefore V_M(\phi, v) = 0$ for all v s.t. Rwv

$\therefore V_M(\Diamond\phi, u) = 1$ for all u s.t. Rwu

$\therefore V_M(\phi, u') = 1$ for some u' s.t. Ruu'

since $Rwu \,\&\, Ruu'$, Rwu'

$\therefore V(\phi, u') = 1 \,\&\, V(\phi, u') = 0$ ✗

if $V_M(\Box\Diamond\phi, w) = 0$, $V_M(\Diamond\phi, w) = 1$

$V_M(\phi, v) = 1$ for some v s.t. Rwv

$V_M(\Diamond\phi, u) = 0$ for some u s.t. Rwu

$V_M(\phi, u') = 0$ for all u' s.t. Ruu'

$Rwu \,\&\, Ruu' \Rightarrow Rwu'$

$\therefore V_M(\phi, v) = 1 \,\&\, V_M(\phi, v) = 0$ ✗

Case 4: let $\bigcirc\phi \equiv_{S5} \Diamond\phi$ & append $\Diamond$ to $\bigcirc\phi$

claim: $\vdash_{S5} \Diamond\Diamond\phi \leftrightarrow \Diamond\phi$

AFC $V_M(\Diamond\Diamond\phi \leftrightarrow \Diamond\phi, w) = 0$

$\therefore V_M(\Diamond\Diamond\phi, w) \neq V_M(\Diamond\phi, w)$

if $V_M(\Diamond\Diamond\phi, w) = 1$ & $V_M(\Diamond\phi, w) = 0$

$V_M(\phi, v) = 0$ for all v s.t. Rwv

$V_M(\Diamond\phi, u) = 1$ for some u s.t. Rwu

$\therefore V_M(\phi, u') = 1$ for some u' s.t. Ruu'

Given Rwu & Ruu' $\Rightarrow$ Rwu'

$\therefore V_M(\phi, u') = 1$ & $V_M(\phi, u') = 0$ $\times$

if $V_M(\Diamond\Diamond\phi, w) = 0$ & $V_M(\Diamond\phi, w) = 1$

$\therefore V_M(\phi, v) = 1$ for some v s.t. Rwv

$\therefore V_M(\Diamond\phi, u) = 0$ for all u s.t. Rwu

$\therefore V_M(\phi, u') = 0$ for all u' s.t. Ruu'

Since $Rwu \Rightarrow Ruw$

Ruw & $Rwv \Rightarrow Ruv$ i.e. v is accessible from u

$\therefore V_M(\phi, v) = 1$ & $V_M(\phi, v) = 0$ $\times$

Therefore, even when $C(\psi) = n+1$, if ϕ has only 2 modalities ie. $\Box\phi, \Diamond\phi$,

then $\Box O\phi, \Diamond O\phi$ is equivalent to either $\Box\phi, \Diamond\phi$ for any O ✓

$\therefore$ There are exactly two modalities expressed by strings in S5

iii. There are exactly 6 modalities expressed by strings in S4.

We first find 6 different modalities expressed by strings in S4 and show that at some complexity n, it will never be more than 6 modalities

1. $\Box\phi$ 2. $\Diamond\phi$

3. $\Diamond\Box\phi$ 4. $\Box\Diamond\phi$

5. $\Box\Diamond\Box\phi$ 6. $\Diamond\Box\Diamond\phi$

Step 1: show that these modalities are not equivalent

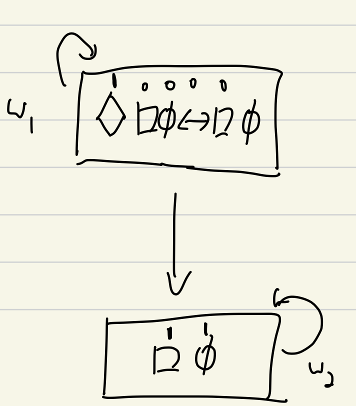
$\nvDash_{S4} \Diamond\Box\phi \leftrightarrow \Box\phi$

$V_M(\Diamond\Box\phi, w) = 1$

$V_M(\Box\phi, v) = 1$, for some v s.t. Rwv

$V_M(\phi, u) = 1$, for all u s.t. Rvu

$V_M(\Box\phi, w) = 0$, $V_M(\phi, u') = 0$ for some u' s.t. Rwu'

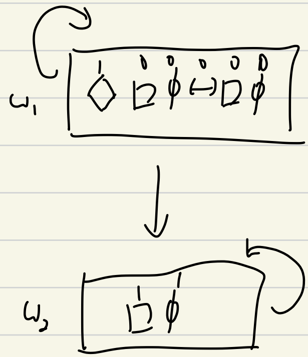


$W = \{w_1, w_2\}$

$R = \{\langle w_1, w_1\rangle, \langle w_1, w_2\rangle, \langle w_2, w_2\rangle\}$

$V_M(\phi, w_1) = 0$, $V_M(\phi, w_2) = 1$

$\nvDash_{S4} \Diamond\Box\phi \leftrightarrow \Box\phi$

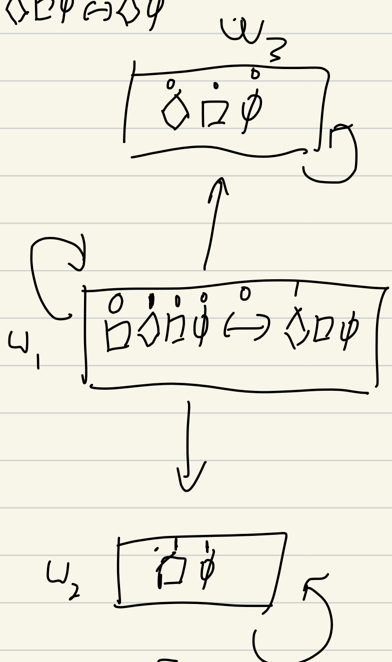


$W = \{w_1, w_2\}$, $R = \{\langle w_1, w_1\rangle, \langle w_2, w_2\rangle, \langle w_1, w_2\rangle\}$

$V_M(\phi, w_1) = 0$, $V_M(\phi, w_2) = 1$

it is obvious that $\nvDash_{S4} \Box\Diamond\phi \leftrightarrow \Box\phi$ & $\nvDash_{S4} \Diamond\Box\phi \leftrightarrow \Diamond\phi$

$\nvDash_{S4} \Box\Diamond\Box\phi \leftrightarrow \Diamond\Box\phi$



$V_M(\Box\Diamond\Box\phi, w) = 0$, $V_M(\Diamond\Box\phi, w) = 1$

$V_M(\Diamond\Box\phi, v) = 0$ for some v s.t. Rwv

$V_M(\Box\phi, u) = 0$ for all u s.t. Rvu

$V_M(\phi, u') = 0$ for some u' s.t. Ruu'

$V_M(\Box\phi, v') = 1$ for some v' s.t. Rwv'

$V_M(\phi, u'') = 1$ for all u'' s.t. $Rv'u''$

$V_M(\phi, w_1) = 0$, $V_M(\phi, w_2) = 1$

$W = \{w_1, w_2\}$

$R = \{\langle w_1, w_1\rangle, \langle w_2, w_2\rangle, \langle w_1, w_2\rangle\}$

$\nvdash_{S4} \Diamond\Box\Diamond\phi \leftrightarrow \Box\Diamond\phi$

AFC $V_M(\Diamond\Box\Diamond\phi \leftrightarrow \Box\Diamond\phi, w)=0$

suppose $V_M(\Diamond\Box\Diamond\phi, w)=1$

$V_M(\Box\Diamond\phi, w)=0$

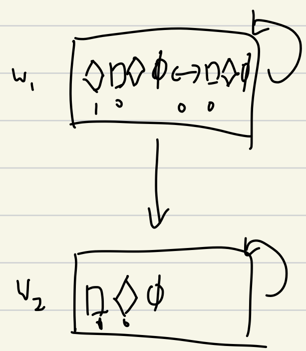
$V_M(\Box\Diamond\phi, v)=1$, for some v s.t. Rwv

$V_M(\Diamond\phi, u)=0$ for some u s.t. Rwu

$V_M(\phi, v')=0$ for all v' s.t. Rvv'

$V_M(\Diamond\phi, u')=1$ for all u' s.t. Rvu'

$V_M(\phi, u'')=1$ for some u'' s.t. $Ru'u''$



We'll see in class

I don't really know how to prove it all out

$\Rightarrow$ I'd expect it would be by induction to show that any formula would terminate at one of these wffs

QS (a) Claim: The no. of occurrences in a PL sentence is twice occurrence of $\to$

let $C(\phi)$ denote no. of connectives

Base case: $C(\phi) = 0$,

then trivially, wff has

0 parantheses

IH: if $C(\phi) = n$, and the no of parantheses is twice the number of $\to$ in ϕ, then for ψ s.t. $C(\psi) = n+1$, the no of parantheses is twice no. of $\to$

Case 1: $\psi := \sim\phi$

Then, since the number of parantheses & $\to$ stay the same, by assumption no. of () is double $\to$

Case 2: $\psi := (\phi \to \chi)$

Then, let the no. of $\to$ in ϕ be some arbitary $k \leq n$

This part of the argument is wrong $\longrightarrow$ $\times$ Now, the no. of $\to$ in ψ is $k+1$. By hypothesis, no. of parenthesis is $2k$ in ϕ, and is $2(k+1)$ in ψ. $\therefore$ no. of ( ) is double no. of $\to$

(b) Claim: for any ϕ with no occurrence of negation, $V_{I^+}(\phi) = 1$

BC: $C(\phi) = 0$, so $\phi := \alpha$. By def $I^+(\alpha) = 1$

ψ does not have to be a sentence letter, ask me in class

IH: if $C(\phi) = n$ & $V_{I^+}(\phi) = 1$, then $V_{I^+}(\phi \circ \psi) = 1$ where $C(\phi \circ \psi) = n+1$ with no occurrence of negation

We know that if $C(\phi \circ \psi) = n+1$ and $C(\phi) = n$, then ψ is some sentence letter.

$\therefore$ $I^+(\psi) = 1$, therefore $V_{I^+}(\psi) = 1$ (*)

By assumption & (*), $V_{I^+}(\phi) = 1$ and $V_{I^+}(\psi) = 1$

so, $V_{I^+}(\phi \wedge \psi) = 1$

$V_{I^+}(\phi \vee \psi) = 1$

$V_{I^+}(\phi \rightarrow \psi) = 1$

$V_{I^+}(\phi \leftrightarrow \psi) = 1$

$\therefore$ IH established $\square$

(c) Base Case: $C(a) = 0$. Then $V_I(a) = 1$ and $V_{I'}(a) = 0$

IH: if $C(\phi) = n$, where ϕ contains at most occurrence of each sentence letter and $\exists I, I'$ s.t. $V_I(\phi) = 1$ and $V_{I'}(\phi) = 0$, then for $C(\psi) = n+1$ where ψ contains at most one sentence letter, $\exists I, I'$ s.t. $V_I(\psi) = n$ and $V_{I'}(\psi) = 0$

Case 1: $\psi ::= \sim\phi$

Then, by assumption since $\exists I, I'$ s.t. $V_I(\phi) = 1$, $V_{I'}(\phi) = 0$

$\therefore V_I(\sim\phi) = 0$, $V_{I'}(\sim\phi) = 1$

Case 2: $\psi ::= \phi \rightarrow \varphi$

Now, φ is a sentence letter not occurring in ϕ, so the base case applies i.e. $V_I(\varphi) = 1$ and $V_{I'}(\varphi) = 0$

Now, since $\text{Senlett}(\phi) \cap \text{Senlette}(\varphi) = \emptyset$,

we can choose the same I' as $V_{I'}(\varphi) = 0$ s.t.

$V_{I'}(\phi) = 1$

$\therefore V_{I'}(\phi \rightarrow \varphi) = 0$

Likewise, we can choose the same I as $V_I(\varphi) = 1$ s.t.

$V_I(\phi \to \varphi) = 1$ when $V_I(\varphi) = 1$

The inductive hypothesis liscences the last two steps since we know that such an I s.t. $V_I(\varphi) = 1$ exists.

✓

Since if ϕ contains at most one occurrence of each sentence letter then there is an I s.t. $V_I(\phi) = 1$ and an I s.t. $V_I(\phi) = 0$, then by the def of validity, $\nvDash_{PL} \phi$ since there is some I s.t. $V_I(\phi) = 0$

Q6. LfP ex 3.7 Show that there are no Kleene valid wffs

i.e. $\nvDash_K \phi$

We prove a stronger claim that for a I that assigns # to all sentence letters, then for any ϕ, $V_I(\phi) = \#$

Base Case: for any α, $C(\alpha) = 0$

Then by def, $I(\alpha) = \#$

IH: if for all φ s.t. $C(\varphi) < n$, $V_I(\varphi) = \#$ where I assigns # to all sentence letters then for ψ s.t. $C(\psi) = n$, then $V_I(\psi) = \#$

Case 1: $\psi := \sim\phi$

By assumption, $V_I(\phi) = \#$, $V_I(\sim\phi) = \#$

Case 2: $\psi := \phi \wedge \varphi$

By assumption, $V_I(\phi) = V_I(\varphi) = \#$ $\therefore V_I(\phi \wedge \varphi) = \#$

Case 3: $\psi := \phi \vee \varphi$

By assumption, $V_I(\phi) = V_I(\varphi) = \#$

$\therefore V_I(\phi \vee \varphi) = \#$

It is sufficient to only do case 1 and case 4

or 1 and 3

or 1 and 2

Case 4: $\psi := \phi \to \varphi$

By assumption, $V_I(\phi) = V_I(\varphi) = \#$

$\therefore V_I(\phi \to \varphi) = \#$

Case 5: $\psi := \phi \leftrightarrow \varphi$

By assumption, $V_I(\phi) = V_I(\varphi) = \#$

$\therefore V_I(\phi \leftrightarrow \varphi) = \#$

So, for any ϕ, $V_I(\phi) = \#$

$\therefore \not\models_K \phi$

LfP ex 3.8. I^+ refines I, if $I(\alpha) = 1$, then $I^+(\alpha) = 1$

if $I(\alpha) = 0$, then $I^+(\alpha) = 0$

Claim is that I^+ preserves the classical truth values of Kleene truth table

We can prove this by induction.

Base Case: Consider any α s.t. $C(\alpha) = 0$ by def, if $I(\alpha) = 1$, then $I^+(\alpha) = 1$

if $I(\alpha) = 0$, then $I^+(\alpha) = 0$

IH: if I^+ preserves the classical truth values of ϕ for $C(\phi) < n$, then it does so for ψ when $C(\psi) = n$

Case 1: $\psi := \sim\phi$

By def, if $V_I(\sim\phi)=0$, $V_I(\phi)=1$,

$V_{I^+}(\phi)=1$, $V_{I^+}(\sim\phi)=0$

Again, only doing 2 cases would make the proof a lot shorter

if $V_I(\sim\phi)=1$, then $V_I(\phi)=0$,

then $V_{I^+}(\phi)=0$, then $V_{I^+}(\sim\phi)=1$

if $V_I(\phi)=\#$, then $V_I(\sim\phi)=\#$,

so no classical value to preserve

Case 2: $\psi := \phi\wedge\varphi$

Again trivially, if $V_I(\phi\wedge\varphi)=1$,

$\therefore V_I(\phi)=V_I(\varphi)=1$

$\therefore V_{I^+}(\phi)=V_{I^+}(\varphi)=1$ by IH,

$\therefore V_{I^+}(\phi\wedge\varphi)=1$

if $V_I(\phi\wedge\varphi)=0$, either $V_I(\phi)=0$ or

$V_I(\varphi)=0$

if $V_I(\phi)=0$, $\underbrace{V_{I^+}(\phi)=0}_{\text{by IH}}$ $\therefore V_{I^+}(\phi\wedge\varphi)=0$

if $V_I(\varphi)=0$, $\underbrace{V_{I^+}(\varphi)=0}_{\text{by IH}}$, $\therefore V_{I^+}(\phi\wedge\varphi)=0$

Case 3: $\phi\vee\varphi$

if $V_I(\phi\vee\varphi)=0$, $V_I(\phi)=V_I(\varphi)=0$

$\therefore \underbrace{V_{I^+}(\phi)=V_{I^+}(\varphi)=0}_{\text{by IH}}$ $\therefore V_{I^+}(\phi\vee\varphi)=0$

if $V_I(\phi \vee \varphi) = 1$, $V_I(\phi) = 1$ or $V_I(\varphi) = 1$

$\therefore V_{I^+}(\phi) = 1$ or $V_{I^+}(\varphi) = 1$, by IH

$\therefore V_{I^+}(\phi \vee \varphi) = 1$ or $V_{I^+}(\phi \vee \varphi) = 1$

$\therefore V_{I^+}(\phi \vee \varphi) = 1$

Case 4: $\phi \to \varphi$

if $V_I(\phi \to \varphi) = 1$, then either $V_I(\phi) = 0$ or $V_I(\varphi) = 1$

$\therefore V_{I^+}(\phi) = 0$ or $V_{I^+}(\varphi) = 1$, by IH

$\therefore V_{I^+}(\phi \to \varphi) = 1$ or $V_{I^+}(\phi \to \varphi) = 1$

$\therefore V_{I^+}(\phi \to \varphi) = 1$

if $V_I(\phi \to \varphi) = 0$, $V_I(\phi) = 1$ & $V_I(\varphi) = 0$

Trivially, $V_{I^+}(\phi) = 1$ & $V_{I^+}(\varphi) = 0$, by IH

$\therefore V_{I^+}(\phi \to \varphi) = 0$

✓

Claim established as I^+ (refinement of I) preserves the classical truth values of the Kleene table

LfP ex 3.9 If we use Lukasiewicz's tables instead

consider an I s.t. $V_I(\phi \to \psi) = 1$

where $V_I(\phi) = \#$ and $V_I(\psi) = \#$

Suppose that we have a refinement I^+

s.t. $V_{I^+}(\phi) = \begin{cases} 0 \text{ if } V_I(\phi) = 0 \\ 1 \text{ otherwise} \end{cases}$

and

$V_{I^+}(\psi) = \begin{cases} 1 \text{ if } V_I(\psi) = 0 \\ 0 \text{ otherwise} \end{cases}$

then $V_{I^+}(\phi) = 1$ & $V_{I^+}(\psi) = 0$ ✓

$\therefore V_{I^+}(\phi \to \psi) = 0$